

1. DBSCAN Core Formulas & Definitions

DBSCAN clusters data based on density using two parameters:

ϵ : Neighborhood radius

MinPts: Minimum neighbors to form a core point

Key Definitions:

Term	Formula/Definition
ϵ -neighborhood	$N_\epsilon(p) = \{q \in D \mid dist(p,q) \leq \epsilon\}$
Core Point	$N_\epsilon(p) \geq MinPts$
Directly Density-Reachable	Point q is directly reachable from p if: p is core point and $q \in N_\epsilon(p)$
Cluster	A maximal set of density-connected points
Noise	Points not in any cluster (labeled -1)

2. Numeric Example

Dataset (6 points from):

Cluster 1: A(1, 1), B(1, 2), C(2, 1)

Cluster 2: D(6, 6), E(6, 7), F(7, 6)

Parameters:

$\epsilon = 1.5$

$MinPts = 3$

Step-by-Step Calculation:

1. Check A(1,1):

- $dist(A,B) = \sqrt{(1-1)^2 + (2-1)^2} = 1.0 \leq 1.5 \rightarrow \text{include}$
- $dist(A,C) = \sqrt{(2-1)^2 + (1-1)^2} = 1.0 \leq 1.5 \rightarrow \text{include}$
- $dist(A,others) > 4.0 \rightarrow \text{exclude}$
- $N_\epsilon(A) = B, C \rightarrow N_\epsilon = 2 < 3 \rightarrow \text{Not core}$

2. **Check B(1,2):**
 - $N_\epsilon(B) = A, C$ (distances: $A = 1.0, C = \sqrt{2} \approx 1.41 \leq 1.5$)
 - $N_\epsilon(B) = 2 < 3 \rightarrow$ **Not core**
3. **Check C(2,1):**
 - $N_\epsilon(C) = A, B \rightarrow N_\epsilon = 2 < 3 \rightarrow$ **Not core**
4. **Points D, E, F:**
 - All have $N_\epsilon = 2$ (e.g., $N_\epsilon(D) = E, F$) $\rightarrow$ **No core points**

Result:

No clusters $\rightarrow$ All points labeled **noise (-1)**

Why? $N_\epsilon = 2 < \text{MinPts}(3)$ for all points.

Fix Parameters (as in):

Set $\epsilon = 1.6$ or $\text{MinPts} = 2$:

B and E become **core points** ($N_\epsilon \geq 2$).

Clusters form: {A,B,C} and {D,E,F}.

3. Algorithm Pseudocode ()

```

labels = [-1] * len(D) # Initialize all as noise (-1)
cluster_id = 0

for each unvisited point p in D:
    if |Nε(p)| ≥ MinPts: # p is core point
        cluster_id += 1
        Expand cluster from p:
            - Add p's neighbors to cluster
            - Recursively add density-reachable points
    else:
        mark p as noise (-1)

```

4. Parameter Selection Tips

1. ϵ : Use a *k-distance plot* (find "elbow" where distance spikes).
2. **MinPts**: Start with *dimensions + 1* (e.g., 3 for 2D data).
3. **Debugging**:
 - All noise? $\rightarrow$ Decrease ϵ or decrease MinPts.
 - One giant cluster? $\rightarrow$ Increase ϵ or increase MinPts.

Key Insight: DBSCAN excels at finding irregular shapes () but fails with uniform density clusters (as shown above). Always validate parameters visually or with metrics like silhouette score.

This explanation uses formal definitions from and practical steps from to demonstrate DBSCAN's strengths/limitations. The example proves that parameter choice directly dictates clustering success.